Simulation Platform & Autonomous Framework for Next-Generation Medical Robotics

Major Technical Project (MTP) — Mid-Evaluation

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Motivation: The Healthcare AI Dilemma

Why Real-World Data Fails

- Modern AI requires **thousands of diverse trials** to master contact-rich interaction.
- Collecting physical patient datasets is:
  - **High Risk:** Cannot risk patient safety.
  - **Resource Intensive:** Clinical scans are costly and time-limited.
  - **Hard to Standardize:** Anatomical variation across patients.
- **Physics-faithful simulation is the mandatory pre-condition for medical robot learning.**

Ultrasound: The Benchmark Task

- Ultrasound is non-invasive and radiation-free.
- However, image acquisition is **strictly coupled to physical contact**:
  - Sub-optimal angle $\rightarrow$ **acoustic shadow / blackout**.
  - Insufficient normal force $\rightarrow$ **acoustic decoupling**.
  - Excessive pressure $\rightarrow$ **tissue deformation / pain**.
- Requires tight, simultaneous coordination of **probe pose, contact force, and image interpretation**.

Core Objectives

Objective 1: Simulation Platform

- **Contact-Rich Dynamics:** Model rigid manipulator and soft-tissue contact mechanics at high frequency.
- **Physical Acoustic Synthesis:** Real-time ultrasound generation simulating wave attenuation, reflections, and speckle.
- **Clinical Suite Integration:** Zero-middleware binary bridge into **3D Slicer** for medical visualization.
- **Virtual Safety Fixtures:** Enforcing keep-out safety barriers directly in task space.

Objective 2: Learning & Benchmarking

- **Active 3D Reconstruction:** Formulate Next-Best-View (NBV) planning to explore unobserved anatomy and resolve shadows.
- **Policy Exploration:** Investigate continuous action representations (e.g., flow matching and diffusion policies).
- **AI Benchmarking Testbed:** Create standardized simulation tasks for evaluating data efficiency and sim-to-real readiness.

Literature Survey (1/2): Robotic Ultrasound & Simulators

SonoGym (Ao et al., 2025)

- **Overview:** Benchmark for robotic ultrasound tracking using clinical CT volumes.
- **Core Strengths:** Fast PyBullet environment for training 2D probe visual servoing.
- **Key Limitations:**
  - **Static Anatomy:** Patient model is rigid; zero tissue deformation under probe pressure.
  - **Acoustic Realism:** Uses 2D GANs without physical ray acoustics (no wave attenuation or bone shadows).
  - **Task Scope:** Confined to 2D image tracking; lacks 3D volumetric reconstruction.

Slicer Bridges (AMBF, SlicerROS2)

- **Overview:** Frameworks bridging physics simulators to 3D Slicer for medical visualization.
- **Core Strengths:** Demonstrates the clinical value of 3D Slicer as an open-source surgical UI.
- **Key Limitations:**
  - **Middleware Complexity:** Relies on heavy ROS/ROS2 communication pipelines.
  - **Serialization Latency:** Multi-process ROS serialization introduces latency and jitter during high-rate control.
  - **Setup Overhead:** Requires multi-node Linux daemons.

Literature Survey (2/2): Surgical Learning & World Models

Surgical VLAs: SRT & SRT-H (2024)

- **Overview:** Vision-Language-Action models and diffusion policies for surgical manipulation.
- **Key Takeaways:**
  - Demonstrates that action chunking and language conditioning yield smooth, reactive motions.
  - High-level language commands enable intuitive supervisory control.
- **Key Limitations:**
  - Tested exclusively on rigid manipulation (peg-transfer and block tasks).
  - Lacks continuous physical contact compliance.

Surgical World Models (Cosmos-H, SAW)

- **Overview:** Video diffusion world models predicting future surgical scenes conditioned on robot actions.
- **Key Takeaways:**
  - Demonstrates the ability of generative models to emulate procedural workflows.
- **Key Limitations:**
  - **Geometric Hallucination:** Generative video models hallucinate anatomical boundaries without rigid physical constraints.
  - **Computational Burden:** High inference latency.

What We Are Doing Differently Than Baselines

Slicer Surgical Bridge

vs. ROS2 / Custom GUIs

- **Direct OpenIGTLink:** Medical-standard binary TCP/IP sockets; zero-ROS middleware bloat.
- **Multi-Rate Timing:** Decouples high-frequency physics from 60 FPS clinical rendering.
- **Virtual Fixtures:** Keep-out safety boundaries enforced in task space.

GPU Acoustic Slicing

vs. 2D GANs (SonoGym)

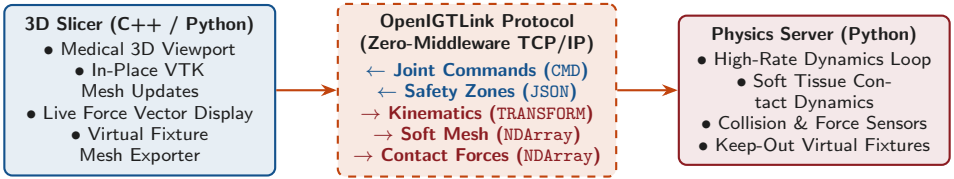
- **Physical Ray Modeling:** Simulates exponential attenuation, boundary reflections, and speckle noise.
- **Zero Hallucination:** Strictly grounded in CT Hounsfield density tensors.
- **Generalizable:** Evaluates zero-shot on any patient CT without GAN retraining.

3D Volumetric Scanning

vs. 2D Tracking / Rasters

- **3D Voxel Accumulation:** Fuses 2D sweeps into a continuous volumetric grid with confidence maps.
- **Active NBV Planning:** Formulates Next-Best-View exploration to resolve acoustic shadows.
- **Volumetric Goal:** Focuses on full organ reconstruction, not just 2D plane tracking.

Work Accomplished: Slicer-PyBullet Surgical Bridge



Decoupled Timing

- Physics steps at high frequency for stable contact dynamics.
- Slicer visualizes at 60 FPS.
- Rendering latency does not degrade contact stability.

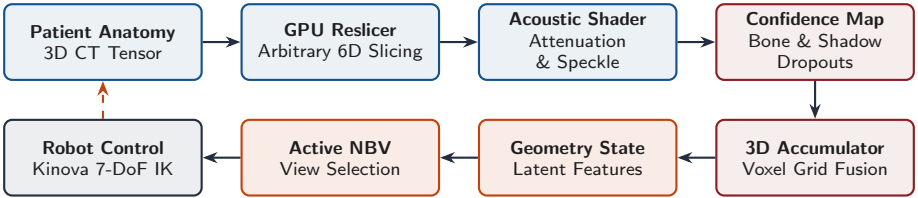
In-Place Memory Updates

- Mutates raw VTK point buffers directly in memory.
- Eliminates object destruction / re-allocation overhead.
- Designed for smooth, interactive visualization.

Virtual Safety Fixtures

- Critical anatomical risk zones defined in 3D Slicer.
- Exported as physical keep-out barriers in PyBullet.
- Prevents probe penetration into sensitive structures.

Work Accomplished: 3D Ultrasound Reconstruction Pipeline



GPU Acoustic Physics

- PyTorch tensor reslicing via `grid_sample`.
- Models exponential beam attenuation.
- Incorporates Rayleigh speckle noise.
- Discards oblique uncoupled rays.

3D Voxel Fusion

- Transforms local 2D scans to global 3D coordinates using tracked probe poses.
- Applies Gaussian spatial fusion.
- Maintains dynamic uncertainty map across voxels.

Active Exploration

- Formulated to prioritize unobserved voxels with high uncertainty.
- Aims to adaptively guide probe trajectory around acoustic obstacles.
- Designed to address scan redundancy.

Autonomous Scanning: Formulating Active View Selection

Limitations of Blind Raster Sweeps

- Baselines typically rely on **open-loop zigzag/raster sweeps**:
  - Blindly traverses a planar bounding box.
  - Misses out-of-plane features, steep bone facets, and occluded contours.
  - Wastes scan time re-sweeping already reconstructed regions.
- Real human anatomy is 3D, non-planar, and acoustically complex.

Active Exploration Formulation

- **Submodular Objective Design:**
  - Re-scanning an already covered voxel yields diminishing information gain.
  - Formulated to balance new surface coverage against robot motion penalty.
- **Expected Benefits:**
  - Designed to steer probe tilt around acoustic obstacles (e.g., ribs).
  - Expected to reduce scan duration and eliminate shadow voids.
- **Status:** Formulation implemented in simulation; systematic testing against baselines planned in future work.

Technical Understanding: Three Core Decisions

Decision 1: Transport

Why OpenIGTLink over ROS?

- Native clinical standard supported by 3D Slicer.
- Avoids multi-node ROS overhead and daemon setup on medical workstations.
- Direct TCP binary streaming provides low jitter and minimal communication latency.

Decision 2: Ultrasound

Why GPU Physics over GANs?

- 2D GANs (e.g., SonoGym) hallucinate features and lack ray acoustics.
- Our GPU PyTorch shader strictly enforces wave attenuation and speckle physics.
- Zero-shot generalization to new patient CTs without model retraining.

Decision 3: Planning

Why Active Planning?

- Standard RL assumes simple additive step rewards.
- Volumetric coverage is inherently history-dependent.
- Formulating submodular diminishing returns guides purposeful exploration.

Future Direction (1/2): Testing & Interactive Control

1. Simulation Pipeline Testing & Verification

- **Multi-Rate Timing Stability:**
  - Stress-test the physics loop under rapid probe contact force transitions and varied penetration depths.
  - Verify collision stability across varied tissue stiffness parameters.
- **GPU Reslicing Profiling:**
  - Profile slicing and acoustic ray shader throughput across diverse patient CT volume sizes.
- **OpenIGTLink Streaming Integrity:**
  - Verify continuous packet transmission during prolonged sessions.

2. Interactive Trajectory Initialization

- **Clinician-in-the-Loop Setup:**
  - Connect 3D Slicer interactive markups (fiducial points and bounding regions) directly to the simulation.
  - Clinicians can delineate a region of interest (e.g., lumbar vertebra) directly on the 3D viewport.
- **Kinematic Goal Resolution:**
  - PyBullet inverse kinematics (IK) automatically resolves clinician markups into initial collision-free robot poses.

Future Direction (2/2): Policy Training & AI Benchmarking

3. AI Policy Training in Simulation

- **Generative Action Chunking:**
  - Train Continuous Flow Matching (CFM) and diffusion policies for smooth 6-DoF probe velocity generation.
- **Active Exploration Optimization:**
  - Train policies using submodular information-gain rewards to dynamically avoid acoustic shadows.
- **Multi-Patient Training:**
  - Train across multiple patient abdominal/spinal CT volumes to promote generalization.

4. Comparative Benchmarking Suite

- **Systematic Baseline Comparison:**
  - Empirically evaluate active planning against standard raster/zigzag sweeps in simulation.
- **Evaluation Metrics (Expected):**
  - Reconstructed volume completeness percentage.
  - Acoustic shadow void reduction.
  - Total trajectory length and scan duration.
- **VLA Observation Testbed:**
  - Structure observation spaces (B-mode ultrasound, contact force history, text prompts) for surgical VLA benchmarking.

Conclusion & Discussion

Summary of Work Accomplished

- **High-Performance Slicer Surgical Bridge:** Real-time integration between PyBullet physics and 3D Slicer via OpenIGTLink with in-place VTK updates and virtual safety fixtures.
- **Autonomous 3D Ultrasound Framework:** GPU-accelerated acoustic physics reslicing, confidence-weighted 3D voxel accumulation, and active view selection formulation.
- **Foundation for Healthcare AI:** Established a clean, decoupled simulation environment ready for pipeline testing and AI policy benchmarking.

Thank You!

Questions & Discussion

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